

Algorithm Design and Analysis

Fernando I. Mondragón Vega

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Chapter 1

Some Representative Problems [1]

1.1 Solved Exercises

Exercise 1

Consider a town with n men and n women seeking to get married to one another. Each man has a preference list that ranks all the women, and each woman has a preference list that ranks all the men. The set of all $2n$ people is divided into two categories: good people and bad people. Suppose that for some number k , $1 \leq k \leq n - 1$, there are k good men and k good women; thus there are $n - k$ bad men and $n - k$ bad women. Everyone would rather marry any good person than any bad person. Formally, each preference list has the property that it ranks each good person of the opposite gender higher than each bad person of the opposite gender: its first k entries are the good people (of the opposite gender) in some order, and its next $n - k$ are the bad people (of the opposite gender) in some order. Show that in every stable matching, every good man is married to a good woman.

Solution (By contradiction)

Let's suppose we get a stable matching S in which a certain good man m is married to a bad woman w' . That implies there's a certain good woman w that is married to a bad man m' , since there is the same amount of good men as good women (k). That means we have an instability, since m prefers w over w' , and w prefers m over m' . This scenario can only happen in two cases:

1. m never proposed to w . However, this cannot happen, since w is ranked higher than w' in m 's preference list.
2. w rejected m . Once again, this is not possible, since m is ranked higher than m' in w 's preference list.

We have found a contradiction, so m , w' could have never match to begin with.

1.1.1 Exercise 2

We can think about a generalization of the Stable Matching Problem in which certain man-woman pairs are explicitly forbidden. In the case of employers and applicants, we could

imagine that certain applicants simply lack the necessary qualifications or certifications, and so they cannot be employed at certain companies, however desirable they may seem. Using the analogy to marriage between men and women, we have a set M of n men, a set W of n women, and a set $F \subseteq M \times W$ of pairs who are simply not allowed to get married. Each man m ranks all the women w for which $(m, w) \notin F$, and each woman w' ranks all the men m' for which $(m', w') \notin F$. In this more general setting, we say that a matching S is stable if it does not exhibit any of the following types of instability.

Bibliography

- [1] Jon Kleinberg and Éva Tardos. *Algorithm Design*. Addison Wesley, 2006.